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Md Yeasin Arafat Tarek

PhD researcher in the department of Mechanical and Mining Engineering at University of Queensland, with a strong foundation in materials science, chemistry, and advanced characterization techniques. Expertise in superconducting joint technology, which includes material synthesis and laboratory-scale process development for the MRI industry. Interested in magnetic and energy storage materials for industrial applications.

Education

Doctor of Philosophy
(Oct 2022 - Present)

The University of
Queensland

**Master of Science in
Chemistry**
(Nov 2018 - June 2021)

Bangladesh University
of Engineering and
Technology (BUET)

**Bachelor of Science in
Chemistry**
(June 2014 - June 2018)

Khulna University

Referee

Md Shahriar Al Hossain
Associate Professor
md.hossain@uq.edu.au

Department of Mechanical
& Mining Engineering
The University of
Queensland

Skills

- Material synthesis and processing, including experimental design, measurement, and characterization.
- Expert in the spark plasma sintering technique.
- Experienced in magnetic and transport measurements of superconducting materials.
- Experience with electrochemical analysis: CV, GCD, EIS
- Experienced with advanced characterization techniques such as SEM, TEM, XRD, XPS, tXRF, Raman, BET, NMR, AAS, FTIR, and GC-MS.

Employment History

Research Fellow (Aug 2021 – Aug 2022)

Bangladesh Council of Scientific & Industrial Research (BCSIR), Dhaka, Bangladesh

Conducted applied research on solar-powered water purification and desalination systems, including material analysis and process development in an ISO-accredited laboratory.

Postgraduate Research Fellow (March 2021 – July 2021)

Chemistry Department, BUET, Dhaka, Bangladesh

Conducted research on biomass-derived Na/K-ion battery materials, including the synthesis, electrochemical evaluation, and advanced characterization techniques.

Experimental Officer (Aug 2020 – Dec 2020)

ESTex, Chemical Engineering Department, BUET, Dhaka, Bangladesh

Performed chemical and elemental analysis of industrial sludge and effluents using instruments such as GC-MS, FTIR, and